

A conjecture on iterated circumcentres

OPG-357

L1 - TRACTABLE

OPEN

Geometry

ORIGINAL PROBLEM STATEMENT

Conjecture Let $p_1, p_2, p_3, \dots$ be a sequence of points in $\mathbb{R}^d$ with the property that for every $i \geq d + 2$, the points $p_{i-1}, p_{i-2}, \dots, p_{i-d-1}$ are distinct, lie on a unique sphere, and further, p_i is the center of this sphere. If this sequence is periodic, must its period be $2d + 4$?

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.
Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>